

Circuit Simulation Lab: 3

Binary Subtractors

Introduction

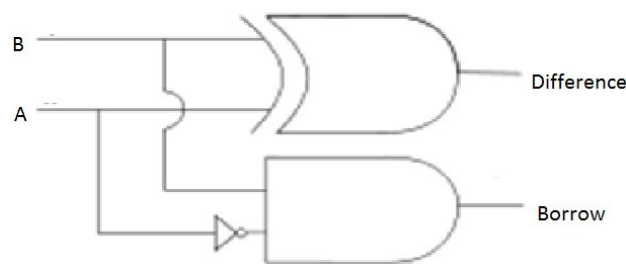
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case half subtractor, and full subtractor.

Software tools Requirement

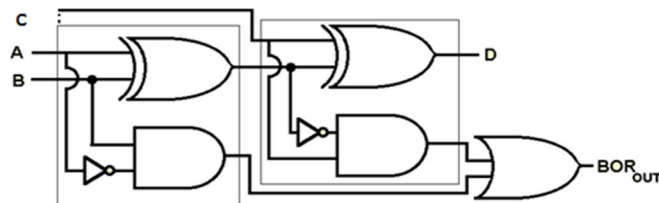
Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



Half Subtractor



Full Subtractor

Describe the following basic adders in Verilog HDL and capture the Waveforms

*Dataflow modeling
Behavior modeling
Structural modeling*